

Mingyue Cui

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Research interests: AI safety, LLM interpretability, secure generative models, and embodied AI

Education

The Hong Kong Polytechnic University

Sep. 2023 – Jun. 2027

B.Sc. in Computer Science; Minor in Finance

Hong Kong

- **CGPA:** 3.77/4.3 | **Honour:** Dean's List
- **Core coursework:** Machine Learning, Artificial Intelligence, Big Data Analysis, Operating Systems, Computer Networking, Computer Organization, Database Security, Software Engineering, Linear Algebra, and Discrete Mathematics.

University of Notre Dame

Jun. 2026 – Jul. 2026

International Summer Undergraduate Research Experience (iSURE)

Notre Dame, IN, USA

Selected Publications

- **Mingyue Cui**, Linghui Shen, and Xingyi Yang. "SAE Interventions are Unreliable: Post-Intervention Recovery of Suppressed Behavior." *arXiv preprint*, 2026. [arXiv:2606.18322](https://arxiv.org/abs/2606.18322)
- Linghui Shen, **Mingyue Cui**, and Xingyi Yang. "BadWorld: Adversarial Attacks on World Models." *Manuscript under review*, 2026. [arXiv:2606.16519](https://arxiv.org/abs/2606.16519)
- Linghui Shen, **Mingyue Cui**, and Xingyi Yang. "DeContext as Defense: Safe Image Editing in Diffusion Transformers." *Under review*. [arXiv:2512.16625](https://arxiv.org/abs/2512.16625)
- ChangLiu, Yuni Lai, **Mingyue Cui**, Cong Tian, Yunyan Zhang, Xian Wu, Kai Zhou, Bin Xiao. "DSPrompt: Dynamic Soft Prompt Defense Against M-RAG Corruption." *Manuscript under review*.

Research Experience

University of Notre Dame

Jun. 2026 – Jul. 2026

Undergraduate Researcher, iSURE / DM2 Group

Notre Dame, IN, USA

Advisors: Prof. Meng Jiang and Zheyuan (Frank) Liu

- Developed **SE-VLA**, distilling safety-shield feedback into $\pi 0.5$ via LoRA and reducing policy-induced safety cost by **35–41%** ($p < 0.01$) with no deployment-time overhead.
- Identified a barrier–cost semantic misalignment in the safety filter and validated it through pre-registered ablations.

The Hong Kong Polytechnic University

Sep. 2025 – Present

Research Assistant, AI Safety and Model Interpretability

Hong Kong

Advisor: Prof. Xingyi Yang

- Led implementation and experiments for **SAE post-intervention recovery** using Python, PyTorch, TransformerLens, SAE Lens, and SAE Bench, demonstrating behavioral recovery across TPP, WMDP-Bio, IOI, and refusal tasks while preserving SAE intervention constraints.
- Co-developed **BadWorld**; implemented self-supervised velocity attacks and trajectory-adaptive bi-level optimization for autoregressive world models, and evaluated robustness across varying action sequences.
- Co-developed **DeContext**; implemented and benchmarked image-protection attacks on FLUX.1-Kontext and Step1X-Edit, evaluating perceptual quality, identity similarity, and face quality.

The Hong Kong Polytechnic University

Oct. 2025 – Present

Research Assistant, LLM and Multimodal RAG Security

Hong Kong

STiL Group; Advisor: Prof. Bin Xiao, Kai Zhou and Dr. Yuni Lai

- Developed a hash-based **MRAG defense** that disperses and detects poisoned evidence via multi-query semantic consistency; benchmarked it against ReliabilityRAG-MIS on PoisonedEye using PyTorch, FAISS, SigLIP and LLaVA.

Industry Experience

ByteDance

Dec. 2025 – Feb. 2026

Machine Learning Intern

Remote

- Built machine-learning and data-analysis workflows for content recommendation on large-scale user data, using Python/SQL for exploratory analysis, feature diagnostics, and experiment evaluation.
- Conducted hypothesis-driven analyses to identify recommendation bottlenecks and translated findings into data- and model-iteration proposals for the team.

PolyU School of Accounting and Finance

Jan. 2025 – Aug. 2025

Student Research Assistant

Hong Kong

- Evaluated LLM-generated educational content, integrated AI tools into digital-business courses, and deployed an interactive Jekyll/GitHub Pages learning site.

Alphabag Limited	Jun. 2025 – Aug. 2025
<i>AI Software Engineer Intern</i>	<i>Hong Kong</i>
Led full-stack development of a Flutter/Firebase mobile app and integrated an LLM-based letter-generation workflow with PDF/Word export for 200+ parents.	
Latent Links Limited	Jun. 2024 – Sep. 2024
<i>AI and Data Marketing Analyst</i>	<i>Hong Kong</i>
Used Python, SQL, and generative-AI tools for competitor analysis, audience segmentation, and data-informed content campaigns.	

Selected Technical Projects

DreamerV3 Reproduction and Policy Optimization	Feb. 2025 – Apr. 2025
Reproduced the DreamerV3 model-based reinforcement-learning pipeline on Atari tasks and explored policy-level modifications to improve training efficiency.	
Human-Computer Interaction: ClassIn Platform Redesign	Sep. 2025 – Nov. 2025
Redesigned key ClassIn interaction flows using HCI principles and developed high-fidelity Figma prototypes with usability-oriented interface and workflow improvements.	
Software Engineering: Jungle Chess Game	Sep. 2025 – Nov. 2025
Developed a command-line Jungle Chess system with modular board-state, rule-engine, and player-interaction components, applying object-oriented software design and collaborative development practices.	
Multi-threaded HTTP Server	Feb. 2025 – Apr. 2025
Implemented a multi-threaded HTTP server with concurrent request handling and optimized networking modules for high-throughput, low-error operation.	
Operating Systems: Smart Parking Scheduling Platform	Jan. 2025 – Apr. 2025
designing dynamic priority scheduling based on resource utilization and time-window constraints	
Student Information Database Security	Sep. 2025 – Nov. 2025
Implemented layered access control, key management, field-level encryption, and secure-query mechanisms for a student-information management system.	

Awards

HKSAR Government Scholarship Fund – Talent Development Scholarship (HK\$10,000)	2026
Honorable Mention, Mathematical Contest in Modeling (MCM)	2026
HKSAR Government Scholarship Fund Award (HK\$80,000)	2025
Dean’s Honours List, The Hong Kong Polytechnic University	2025/26
Seoul National University Exchange Nomination (two university-wide places)	2025
Wong Tit-shing Student Exchange Scholarship Nomination	2025
First Prize, 6th College Students’ Financial Literacy Competition	2025
Poster: “Optimizing Sports Facility Management Using OS-Level Process Scheduling: A Smart Parking System for Sports Venues,” 1st WCSST	2025

Technical Skills

Programming	Python, C/C++, Java, SQL/MySQL, MATLAB, HTML/CSS
ML and research	PyTorch, Hugging Face, LLM/SAE analysis, diffusion models, reinforcement learning, VLA evaluation, adversarial and safety evaluation
Tools	Linux, Git, SLURM/HPC, LaTeX, Jekyll, Flutter, Firebase
Languages	Mandarin Chinese (native); English (professional and academic proficiency)